

Maram Khalid Mohamed

AI Engineer · Multi-Agent Architectures

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OBJECTIVE

AI Engineer with hands-on experience, building machine learning and deep learning solutions across computer vision, natural language processing, and signal processing. Demonstrated success developing production-grade models for healthcare and brain-computer interface applications using PyTorch, TensorFlow, and modern transformer architectures. Motivated to contribute to impactful AI projects that solve real-world problems alongside passionate teams in a high-impact environment.

EDUCATION

Bachelor's in computer science – Major: Artificial Intelligence

Faculty of Computers and Artificial Intelligence, Helwan University

2021 – 2025

- **GPA:** 3.2/4.0
- **Graduation Project: Brain Signals to Text Decoding and Sentiment Classification**
 - **Technologies:** PyTorch, Hugging Face Transformers, T5, BART, EEG Signal Processing, NLP, Deep Learning, NumPy, Pandas, Scikit-learn
 - **Achievement:** Built an end-to-end EEG-to-text decoding pipeline using transformer models (T5, BART) achieving 48% reconstruction accuracy surpassing prior benchmarks and implemented sentiment classification directly from brain signals.

EXPERIENCE

AI Instructor

National Telecommunication Institute (NTI) | Cairo, Egypt

Feb 2025 – Present

- Teaching AI & machine learning fundamentals to early-career engineers.
- Designing hands-on labs covering deep learning, and computer vision.
- Mentoring trainees through capstone projects from problem framing to deployment.

Freelance AI Engineer – Self Employed

Upwork – Khamsat

Feb 2025 – Present

- Developed a computer vision model for a dental clinic focused on object detection of oral disease (caries) from clinical images to assist in diagnosis.
- Built a classification model for diabetic foot ulcer conditions integrated into the “Rafiqi” chatbot, providing patients with support and information related to diabetic foot ulcers.
- Owned end-to-end model lifecycle: data preprocessing, model training, evaluation, and delivery to clients.

Junior AI Engineer – Full Time

Beyond App Groups | Cairo, Egypt

Sep 2025 – Mar 2026

- Designed and deployed production-grade multi-agent AI orchestration systems using LangGraph with hybrid reasoning pipelines combining LLM agents and structured logic execution.
- Developed and integrated AI-powered production features into the MyZesty platform.
- Built production-ready AI services deployed via serverless AWS Lambda pipelines supporting real-world application workflows
- Collaborated with backend and product teams to design and integrate AI services within real-world application environments.

AI Engineer Intern

National Telecommunication Institute (NTI) | Cairo, Egypt

Jul 2025 – Sep 2025

- Designing and implementing full-stack CV pipelines for object detection (YOLOv8, SSD), image segmentation (U-Net, Mask R-CNN), and transformer-based models ViT.
- Optimizing and deploying models using FastAPI, and Docker following modern MLOps best practices.
- Gaining deep technical knowledge in advanced CNN architectures, Vision Transformers, generative models (GANs, Diffusion), and self-supervised learning techniques.

Computer Vision Intern

National Telecommunication Institute (NTI) | Nasr City, Cairo, Egypt

Jan 2025 – Feb 2025

- **Key Project:** Dental Condition Classification System.
- Fine-tuned a ResNet18 model in PyTorch to classify dental conditions including caries and ulcers from image data, achieving 94% accuracy.

AI Instructor (Volunteer)

IEEE & MSP | Helwan University

Oct 2022 – Jun 2024

- Led workshops and training sessions for students in machine learning, deep learning, and Python programming.
- Coordinated technical events and activities to promote innovation, leadership, and skill development.
- Mentored students through building AI projects and participating in technical competitions.

KEY PROJECTS

Agentic Systems

- **Genie Edit** — Developed a multi-agent AI system orchestrating 5+ automated workflows that transform natural-language input into structured editing, cutting the average user task from ~15 manual clicks to a single prompt.

Computer Vision

- **Dental Caries Detection** – Built an object detection model to identify oral disease (caries) from dental images for clinical decision support.
- **Dental Condition Classification** – Fine-tuned ResNet18 in PyTorch on a multi-class dental dataset (caries, ulcers, etc.) achieving 94% accuracy.
- **Diabetic Foot Ulcer Classification** – Developed a CNN-based classifier integrated into the “Rafiqi” chatbot to recognize diabetic foot ulcer conditions.

Natural Language Processing (NLP)

- **Brain Signals to Text Decoding** – Implemented and fine-tuned T5 and BART models on the ZUCO EEG dataset, achieving 48% accuracy and surpassing prior benchmarks.
- **EEG Sentiment Classification** – Designed a sentiment classification pipeline on EEG signals as part of an end-to-end EEG-to-Text framework

Data Science & Machine Learning

- **Air Quality Prediction** – Built a Deep Neural Network (DNN) regression model to predict air pollution levels using temperature, humidity, wind speed, and emissions data.
- **Coevolving ANN-Based Gomoku AI** – Designed an AI agent to play Gomoku at multiple difficulty levels using Artificial Neural Networks combined with Genetic Algorithms.

SKILLS

- **Programming Languages:** Python, C/C++, Java, JavaScript
- **Deep Learning & Libraries:** PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Seaborn, Matplotlib
- **Computer Vision:** YOLO, SSD, U-Net, Pose Estimation, Variational Autoencoders (VAE), ViT
- **Natural Language Processing (NLP):** Sentiment Analysis, T5, BART, BERT, RAG, Text Classification, Web Scraping, Transformers, Text Preprocessing
- **Data Science:** Regression, Classification, Time Series Forecasting, Exploratory Data Analysis (EDA), Data Wrangling
- **Data Engineering & Analytics:** MySQL, Power BI
- **Tools & Technologies:** Git, Docker, FastAPI, Streamlit, Ngrok
- **Languages:** Arabic (native), English (fluent)